

tb_cocotb.v

AUTHORS

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DATES

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INFORMATION

Brief

Test bench wrapper for cocotb

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tb_cocotb

```
module tb_cocotb #(
  parameter
  integer
  ADDRESS_WIDTH
  =
  32,
  parameter
  integer
  BUS_WIDTH
  =
  4,
  parameter
  [ADDRESS_WIDTH-1:0]
  SLAVE_ADDRESS
```

```

    =
    32'h44A20000,
    parameter
    [ADDRESS_WIDTH-1:0]
    SLAVE_REGION
    =
    32'h0000FFFF
)

output
wire
connected,
input
wire
aclk,
input
wire
arstn,
input
wire
[ADDRESS_WIDTH-1:0]
s_axi_araddr,
input
wire
[2:0]
s_axi_arprot,
input
wire
s_axi_arvalid,
output
wire
s_axi_arready,
output
wire
[BUS_WIDTH*8-1:0]
s_axi_rdata,
output
wire
[1:0]
s_axi_rresp,
output
wire
s_axi_rvalid,
input
wire
s_axi_rready,
output
wire
[ADDRESS_WIDTH-1:0]
m_axi_araddr,
output
wire
[2:0]
m_axi_arprot,
output
wire
m_axi_arvalid,
input
wire
m_axi_arready,
input
wire
[BUS_WIDTH*8-1:0]
m_axi_rdata,
input
wire

```

(

```

    [1:0]
m_axi_rresp,
input
wire
m_axi_rvalid,
output
wire
m_axi_rready,
input
wire
    [ADDRESS_WIDTH-1:0]
s_axi_awaddr,
input
wire
    [2:0]
s_axi_awprot,
input
wire
s_axi_awvalid,
output
wire
s_axi_awready,
input
wire
    [BUS_WIDTH*8-1:0]
s_axi_wdata,
input
wire
    [BUS_WIDTH-1:0]
s_axi_wstrb,
input
wire
s_axi_wvalid,
output
wire
s_axi_wready,
output
wire
    [1:0]
s_axi_bresp,
output
wire
s_axi_bvalid,
input
wire
s_axi_bready,
output
wire
    [ADDRESS_WIDTH-1:0]
m_axi_awaddr,
output
wire
    [2:0]
m_axi_awprot,
output
wire
m_axi_awvalid,
input
wire
m_axi_awready,
output
wire
    [BUS_WIDTH*8-1:0]
m_axi_wdata,
output
wire

```

```

[BUS_WIDTH-1:0]
m_axi_wstrb,
output
wire
m_axi_wvalid,
input
wire
m_axi_wready,
input
wire
[1:0]
m_axi_bresp,
input
wire
m_axi_bvalid,
output
wire
m_axi_bready
)

```

AXI lite address recoder for read channel. Write channel lines included for cocotb cores to connect to.

Parameters

ADDRESS_WIDTH parameter integer	Width of the AXI LITE address port in bits.
BUS_WIDTH parameter integer	Width of the AXI LITE bus data port in bytes.
SLAVE_ADDRESS parameter [ADDRESS_WIDTH- 1:0]	Array of Addresses for each slave (0 = slave 0 and so on).
SLAVE_REGION parameter [ADDRESS_WIDTH- 1:0]	Region for the address that is valid for the SLAVE ADDRESS.

Ports

connected output wire	Core has established channel connection
aclk input wire	Input clock
arstn input wire	Input negative reset
s_axi_araddr input wire [ADDRESS_WIDTH- 1:0]	Slave read chanel input address.
s_axi_arprot input wire [2:0]	Slave read chanel input address protection
s_axi_arvalid input wire	Slave read chanel input address is valid
s_axi_arready output wire	Slave read chanel input is ready.
s_axi_rdata output wire [BUS_WIDTH* 8- 1:0]	Slave read chanel input data.
s_axi_rresp output wire [1:0]	Slave read chanel input data response.
s_axi_rvalid output wire	Slave read chanel input data valid
s_axi_rready input wire	Slave read chanel input is ready.
m_axi_araddr output wire [ADDRESS_WIDTH- 1:0]	Master read chanel output address.

m_axi_arprot output wire [2:0]	Master read chanel output address protection
m_axi_arvalid output wire	Master read chanel output address is valid
m_axi_arready input wire	Master read chanel output is ready.
m_axi_rdata input wire [BUS_WIDTH* 8- 1:0]	Master read chanel output data.
m_axi_rresp input wire [1:0]	Master read chanel output data response.
m_axi_rvalid input wire	Master read chanel output data valid
m_axi_rready output wire	Master read chanel output is ready.

INSTANTIATED MODULES

dut

Device under test, axi_lite_read_channel_decoder